Stochastic Processes Semester 1

Tutorial 3 – Solutions

1. (a) i. We have

$$P^2 = \begin{pmatrix} 3/8 & 1/4 & 3/8 \\ 1/2 & 0 & 1/2 \\ 11/16 & 1/8 & 3/16 \end{pmatrix} \qquad P^3 = \begin{pmatrix} 17/32 & 3/16 & 9/32 \\ 3/8 & 1/4 & 3/8 \\ 33/64 & 3/32 & 25/64 \end{pmatrix}.$$

[*Verify that these are stochastic matrices (i.e. all entries non-negative and all rows sum to 1).*]

ii. If the chain is in state 1 at time $n = 5$, the probability that it is in state 0 at time $n = 7$ is

$$\mathbf{P}(X_7 = 0 \,|\, X_5 = 1) = \mathbf{P}(X_2 = 0 \,|\, X_0 = 1) = p_{10}^{(2)} = 1/2.$$

(b) i. Let $\boldsymbol{\mu}^{(n)} = (\mu_i^{(n)})$ where $\mu_i^{(n)} = \mathbf{P}(X_n = i)$. Then

$$\boldsymbol{\mu}^{(3)} = \boldsymbol{\mu}^{(0)} P^3$$

and so, if

$$\boldsymbol{\mu}^{(0)} = (\mathbf{P}(X_0 = 0), \mathbf{P}(X_0 = 1), \mathbf{P}(X_0 = 2)) = (1/3, 1/3, 1/3)$$

we have

$$\boldsymbol{\mu}^{(3)} = (\mathbf{P}(X_3 = 0), \mathbf{P}(X_3 = 1), \mathbf{P}(X_3 = 2)) = (91/192, 34/192, 67/192).$$

[*Verify that these probabilities sum to* 1.]

ii. We further have

$$\begin{aligned} \mathbf{P}(X_0 = i \text{ and } X_3 = 2) &= \mathbf{P}(X_0 = i)\mathbf{P}(X_3 = 2 | X_0 = i) \\ &= \mathbf{P}(X_0 = i) p_{i2}^{(3)} \\ &= \frac{1}{3}(9/32, 3/8, 25/64), \qquad i = 0, 1, 2 \\ &= (18/192, 24/192, 25/192), \qquad i = 0, 1, 2. \end{aligned}$$

and also $\mathbf{P}(X_3 = 2) = 67/192$ (from (i)). Hence

$$\begin{aligned} \mathbf{P}(X_0 = i \,|\, X_3 = 2) &= \frac{\mathbf{P}(X_0 = i \text{ and } X_3 = 2)}{\mathbf{P}(X_3 = 2)} \\ &= (18/67, 24/67, 25/67), \qquad i = 0, 1, 2. \end{aligned}$$

[*Again verify that these probabilities sum to* 1.]

2. (a) The required transition matrix is

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 3/4 & 0 & 1/4 \\ 3/4 & 1/4 & 0 \end{pmatrix}.$$

(b) We have

$$P^3 = \begin{pmatrix} 3/16 & 13/32 & 13/32 \\ 39/64 & 3/16 & 13/64 \\ 39/64 & 13/64 & 3/16 \end{pmatrix}.$$

Labelling the states A, B, C, both the required probabilities are

$$\mathbf{P}(X_3 = C \,|\, X_0 = A) = \mathbf{P}(X_{13} = C \,|\, X_{10} = A) = p_{AC}^{(3)} = 13/32.$$

3. (a) It is easy to check that the required result is correct in the case $n = 1$ (indeed to start the induction it is sufficient to check $n = 0$). Suppose that the result is true for $n = k$. Then

$$\begin{aligned} P^{k+1} &= P^k P \\ &= \begin{pmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{pmatrix} \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix} + \left(-\frac{1}{2}\right)^k \begin{pmatrix} 1/3 & -1/3 \\ -2/3 & 2/3 \end{pmatrix} \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^k \begin{pmatrix} -1/6 & 1/6 \\ 1/3 & -1/3 \end{pmatrix} \\ &= \begin{pmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^{k+1} \begin{pmatrix} 1/3 & -1/3 \\ -2/3 & 2/3 \end{pmatrix} \end{aligned}$$

and so the result is true also for $n = k+1$. Hence the required induction follows.

(b) Since $(-1/2)^n \to 0$ as $n \to \infty$, the required limit result is immediate.

(c) i. If $\mathbf{P}(X_0 = i) = 1/2$ for $i = 1, 2$, then the vector of probabilities

$$\boldsymbol{\mu}^{(n)} = (\mathbf{P}(X_n = 1), \mathbf{P}(X_n = 2))$$

is given by

$$\begin{aligned} \boldsymbol{\mu}^{(n)} &= \boldsymbol{\mu}^{(0)} P^n \\ &= \begin{pmatrix} 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 1/3 & -1/3 \\ -2/3 & 2/3 \end{pmatrix} \\ &= \begin{pmatrix} 2/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} -1/6 & 1/6 \end{pmatrix}. \end{aligned}$$

ii. Now take the limit as $n \to \infty$:

$$\lim_{n\to\infty} \boldsymbol{\mu}^{(n)} = \begin{pmatrix} 2/3 & 1/3 \end{pmatrix}.$$

(d) If $\mathbf{P}(X_0 = 1) = 2/3$ and $\mathbf{P}(X_0 = 2) = 1/3$, then, for all n,

$$\begin{aligned} \boldsymbol{\mu}^{(n)} &= (\mathbf{P}(X_n = 1), \mathbf{P}(X_n = 2)) \\ &= \boldsymbol{\mu}^{(0)} P^n \\ &= \begin{pmatrix} 2/3 & 1/3 \end{pmatrix} \begin{pmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} 2/3 & 1/3 \end{pmatrix} \begin{pmatrix} 1/3 & -1/3 \\ -2/3 & 2/3 \end{pmatrix} \\ &= \begin{pmatrix} 2/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 2/3 & 1/3 \end{pmatrix}, \end{aligned}$$

i.e. the initial distribution is *stationary*.

4. The associated Markov chain has state space $S = (1, 2)$, where 1=dividend, 2=no dividend, and transition matrix

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.6 & 0.4 \end{pmatrix}.$$

(a)

$$P^3 = \begin{pmatrix} 0.861 & 0.139 \\ 0.834 & 0.166 \end{pmatrix}.$$

The required probability is $p_{11}^{(3)} = 0.861$.

(b)

$$P^{11} = \begin{pmatrix} 0.857 & 0.143 \\ 0.857 & 0.143 \end{pmatrix}.$$

The required probability is $p_{11}^{(11)} = 0.857$.

5. (a) All states intercommunicate, and hence the entire state space is a single closed class (for which the vector of absorption probabilities is of course $\mathbf{1} = (1, 1, 1, 1)^T$).

   (b) The classes of intercommunicating states are

Class	Type
$\{1, 2\}$	open
$\{3\}$	open
$\{4\}$	closed (absorbing)

   Note that:

   - $\{4\}$ is closed because $p_{44} = 1$;
   - $\{3\}$ is open because $p_{34} > 0$ and $\{4\}$ is closed, and it is a class on its own because $p_{13} = p_{23} = 0$;
   - $\{1, 2\}$ is open because, say, $p_{13} > 0$, while 1 and 2 are in the same class because $p_{12}p_{21} > 0$.

   Since the state space is finite, and there is a single closed class $\{4\}$, the vector of absorption probabilities associated with this class is again

$$\boldsymbol{\alpha}^{\{4\}} = (\alpha_1^{\{4\}}, \alpha_2^{\{4\}}, \alpha_3^{\{4\}}, \alpha_4^{\{4\}})^T = \mathbf{1} = (1, 1, 1, 1)^T.$$

   [Recall that when the state space is finite, then, for any initial state, the sum of the absorption probabilities over all closed classes is 1.]

   (c) The classes of intercommunicating states are

Class	Type
$\{1, 3, 6\}$	closed
$\{2\}$	open
$\{4\}$	closed (absorbing)
$\{5, 7\}$	open

   Note that:

   - $\{4\}$ is closed because $p_{44} = 1$;
   - $\{1, 3, 6\}$ is closed because

$$p_{11} + p_{13} + p_{16} = p_{31} + p_{33} + p_{36} = p_{61} + p_{63} + p_{66} = 1$$

     and all 3 states intercommunicate since $p_{13}p_{36}p_{61} > 0$;
   - $\{2\}$ is open because, say, $p_{21} > 0$ but it is a class on its own because $p_{i2} = 0$ for all $i = 1, \ldots, 7$;
   - $\{5, 7\}$ is open because, say, $p_{51} > 0$, while 5 and 7 are in the same class because $p_{57}p_{75} > 0$.

   Let $\boldsymbol{\alpha}^{\{4\}}$ be the vector of absorption probabilities in the closed class consisting of the single absorbing state 4. We have $\alpha_1^{\{4\}} = \alpha_3^{\{4\}} = \alpha_6^{\{4\}} = 0$, $\alpha_4^{\{4\}} = 1$, and

$$\begin{aligned}
\alpha_2^{\{4\}} &= \frac{1}{4}\alpha_5^{\{4\}} \\
\alpha_5^{\{4\}} &= \frac{1}{6} + \frac{1}{6}\alpha_5^{\{4\}} + \frac{1}{2}\alpha_7^{\{4\}} \\
\alpha_7^{\{4\}} &= \alpha_5^{\{4\}},
\end{aligned}$$

   giving $\alpha_2^{\{4\}} = 1/8$, $\alpha_5^{\{4\}} = \alpha_7^{\{4\}} = 1/2$.
   Further, since the state space is finite, for any initial state, the sum of the absorption probabilities over all closed classes is 1. Hence the vector of absorption probabilities associated with the closed class $\{1, 3, 6\}$ is

$$\boldsymbol{\alpha}^{\{1,3,6\}} = \mathbf{1} - \boldsymbol{\alpha}^{\{4\}} = (1, 7/8, 1, 0, 1/2, 1, 1/2)^T.$$

6. To check if $\{X_n\}_{n\geq 1}$ is a Markov chain we have to verify two things:

   (a) $\{X_n\}_{n\geq 1}$ has a discrete state space;

   (b) $\{X_n\}_{n\geq 1}$ satisfies the Markov property.

   The first condition is satisfied since X_n can only take on the values of 0, 1 or 2 and hence has a discrete state space. Let us now demonstrate a scenario in which the Markov property fails.

   If $X_{n-1} = 2$ then it follows from the definition that $Y_{n-1} + Y_{n-2} = 2$ and so both $Y_{n-1} = 1$ and $Y_{n-2} = 1$. If in addition to this, $X_n = 1$ then $Y_n = 0$ since

   $$Y_n = X_n - Y_{n-1} = 1 - 1 = 0.$$

   Thus,

   $$\mathbf{P}(X_{n+1} = 2|X_n = 1, X_{n-1} = 2) = \mathbf{P}(X_{n+1} = 2|Y_n = 0) = 0.$$

   On the other hand, if $X_n = 1$, without knowing the value of X_{n-1}, we could have $Y_n = 1$. Thus,

   $$\mathbf{P}(X_{n+1} = 2|X_n = 1) > 0.$$

   Since $\mathbf{P}(X_{n+1} = 2|X_n = 1) \neq \mathbf{P}(X_{n+1} = 2|X_n = 1, X_{n-1} = 2)$, the Markov property does not hold.

   [As an exercise, you can try to calculate $\mathbf{P}(X_{n+1} = 2|X_n = 1)$]